FNCE 30007 DERIVATIVE SECURITIES

Tutorial Solutions

Futures

Question 1
The part with a short position in a futures contract sometimes has options as to the precise asset that will be delivered, where delivery will take place, when delivery will take place, and so on. Do these options increase or decrease the futures price? Explain your reasoning.

Answer
These options make the contract less attractive to the long position and more attractive to the short position. This tends to reduce the futures price.

Question 2
Explain how margins protect investors against the possibility of default.

Answer
A margin is a sum of money deposited by an investor/hedger with their broker. It acts as a guarantee that the investor can cover losses on the futures contract. The balance is adjusted daily to reflect gains and losses. If losses exceed a certain level the investor must deposit a further margin. This makes it less likely they will default.

Question 3
An investor enters into two long July futures contracts on frozen orange juice. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is $6,000 per contract, and the maintenance margin is $4,500 per contract. What price change would lead to a margin call? Under what circumstances could $2,000 be withdrawn from the margin account?

Answer
A margin call occurs if $1500 is lost one 1 contract. This happens if the futures price falls by 10 cents to 150 cents per lb. $2000 can be withdrawn from the margin account if there is a gain on 1 contract of $1000. This will happen if the futures price rises by 6.67 cents to 166.67 cents per lb.

Question 4
Show that, if the futures price of a commodity is greater than the spot price during the delivery period, there is an arbitrage opportunity. Does an arbitrage opportunity exist if the futures price is less than the spot price? Explain your answer.

Answer
If the futures price>spot price at maturity an arbitrageur buys the asset, shorts the futures and makes delivery for immediate profit. If the futures price<spot price there is no similar perfect arbitrage. The arbitrageur can take a long futures position but cannot force immediate delivery of the asset. The decision on when delivery will occur is made by the short party. Nevertheless companies that seek to acquire the asset will take a long position and wait for delivery to be made.

Question 5
On July 1, 2025, a U.S. company enters into a forward contract to buy 10 million British pounds on January 1, 2026. On September 1, 2025, it enters into a forward contract to sell 10 million British pounds on January 1, 2026. Describe the profit or loss the company will make in dollars as a function of the forward exchange rates on July 1, 2025, and September 1, 2025.

Answer
Let F1 and F2 denote the forward exchange rates for contracts entered into on July 1, 2025 and September 1, 2025 and let S denote the spot rate on January 1, 2026. All exchange rates are measured as dollars per pound. The payoff from the first contract is 10(S-F1) million dollars and the payoff from the second contract is 10(F2-S) million dollars. The total payoff is therefore 10(S-F1)+10(F2-S) = 10(F2-F1) million dollars.

Question 6
What do you think would happen if an exchange started trading a contract in which the quality of the underlying asset was incompletely specified?

Answer
The contract would not be a success. Short parties would deliver the cheapest form of the asset which may be totally unacceptable to the long party. Once the quality problem became well known no-one would be prepared to go long. Futures contracts are only feasible when there are rigorous standards defining the quality of the asset. Many futures contracts have failed because of the difficulties associated with defining quality.

Question 7
An airline executive has argued: 'There is no point in our using oil futures. There is just as much chance that the price of oil in the future will be less than the futures price as there is that it will be greater than this price.' Discuss the executive's viewpoint.

Answer
It may well be true that there is just as much chance that the price of oil in the future will be above the futures price as that it will be below the futures price. This means that the use of a futures contract for speculation would be like betting on whether a coin comes up heads or tails. But it might make sense for the airline to use futures for hedging rather than speculation. The futures contract then has the effect of reducing risks. It can be argued that an airline should not expose its shareholders to risks associated with the future price of oil when there are contracts available to hedge the risks.